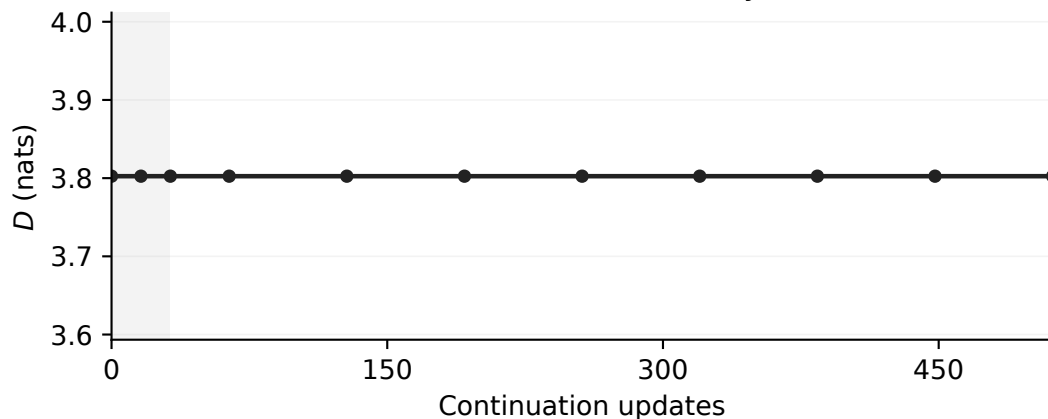
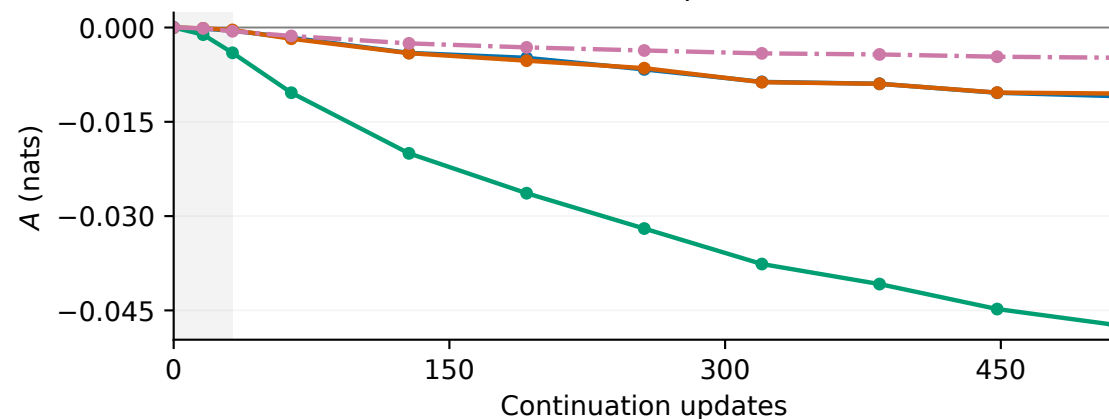


SGD (smaller rate)  
 effective  $h=3e-06$   
 Adam (calibrated)  
 raw  $\eta=2.9e-07$   
 Heavy Ball (smaller rate)  
 effective  $h=3e-06$   
 Head only  
 raw  $\eta=3e-06$

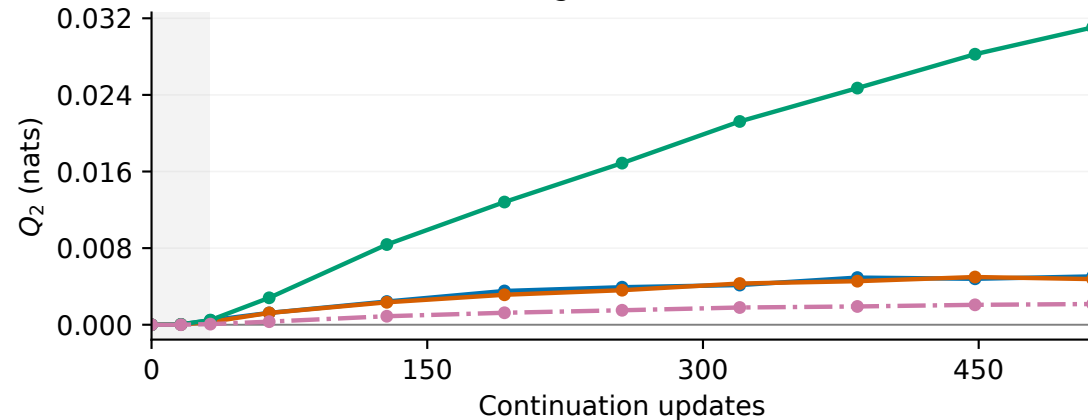
(a) Shared frozen difficulty  $D$



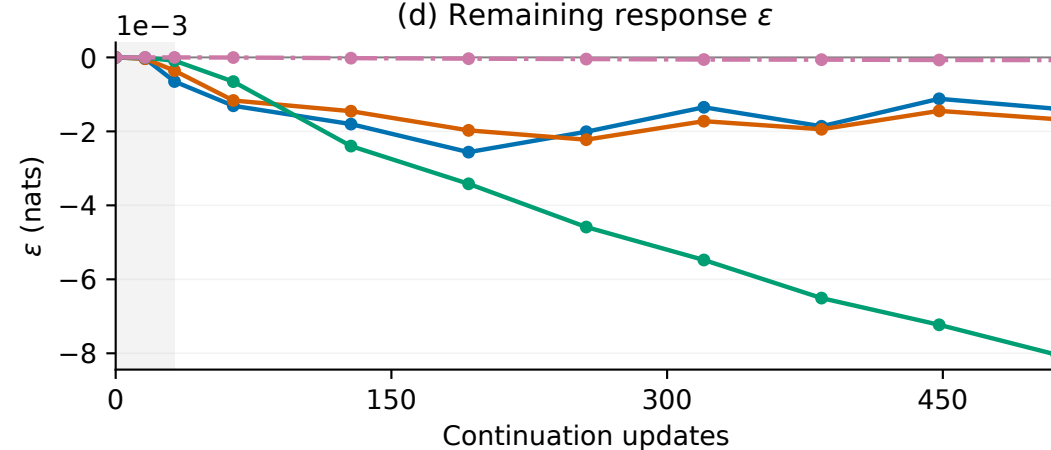
(b) Linear response  $A$



(c) Logit variance  $Q_2$



(d) Remaining response  $\varepsilon$



Dots: fixed held-out 32-document readouts; frozen  $D$  is horizontal.  
 Rate-refinement protocol (adaptive); permutation 1 of 3. Smaller-rate/calibrated settings only; larger-rate results are in the full figure.  
 Heavy Ball raw  $\eta = 0.1h$ ; Adam rate matches one reset-step  $Q_2$ , not its whole trajectory.  
 Grey: 32-update warmup. Shared batch order; separate vertical scales; all nats. Rate labels do not imply stability regimes.